

DSAA 4011 · LAB 4 · FRIDAY 9 OCTOBER 2026

Sampling What We Cannot Sum

Markov Chain Monte Carlo at the Frontier

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Builds on Lecture 4: Sampling replaces an impossible sum

Outline

The walker of Lecture 4

Three troubles, three fixes

The walker as a generator

Conclusion

Outline

► The walker of Lecture 4

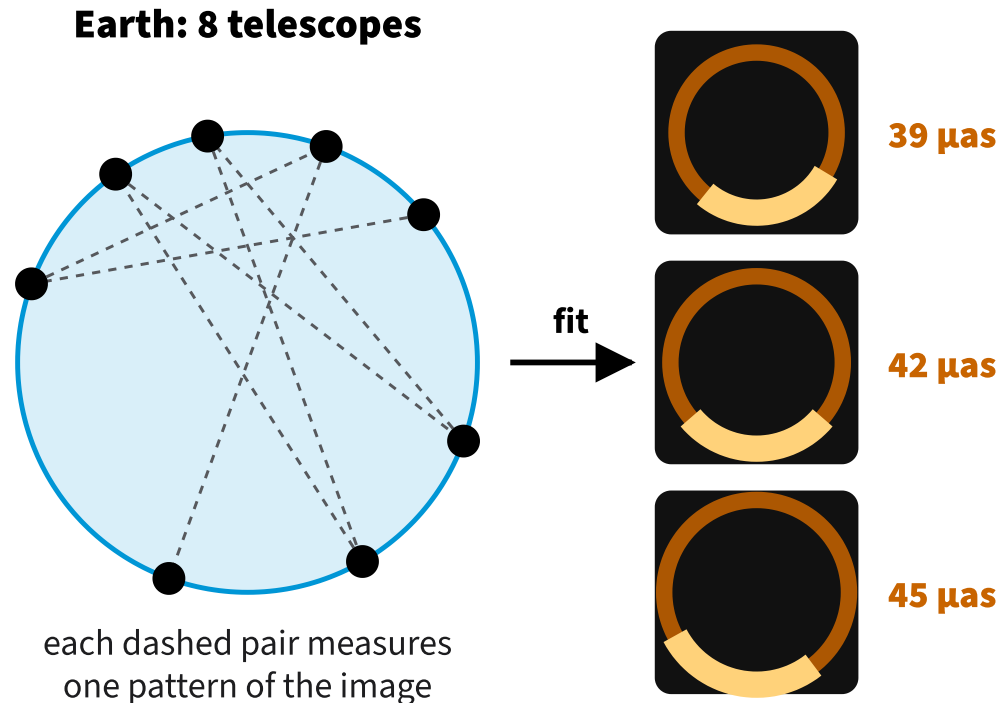
Three troubles, three fixes

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The walker of Lecture 4

April 2019: the first picture of a black hole.



three samples: ring diameter in microarcseconds

the spread of many samples = the error bar

Eight telescopes saw very little: many rings fit (schematic).

Q: Which picture is right?

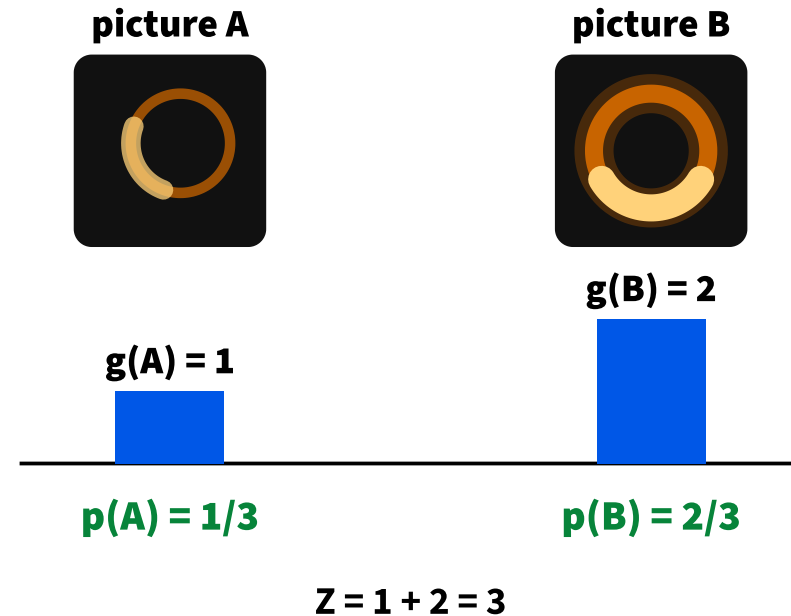
A: Many fit. Draw them all, each as often as it fits.

The walker of Lecture 4

Main idea: draw each picture x as often as it fits.

$$p(x) = \underbrace{g(x)}_{\text{✓ fit: easy}} / \underbrace{\sum}_{\text{× sum over all pictures}}$$

Example Two pictures with fit 1 and 2: draw them $\frac{1}{3}$ and $\frac{2}{3}$ of the time.



The better fit is drawn twice as often.

The walker of Lecture 4

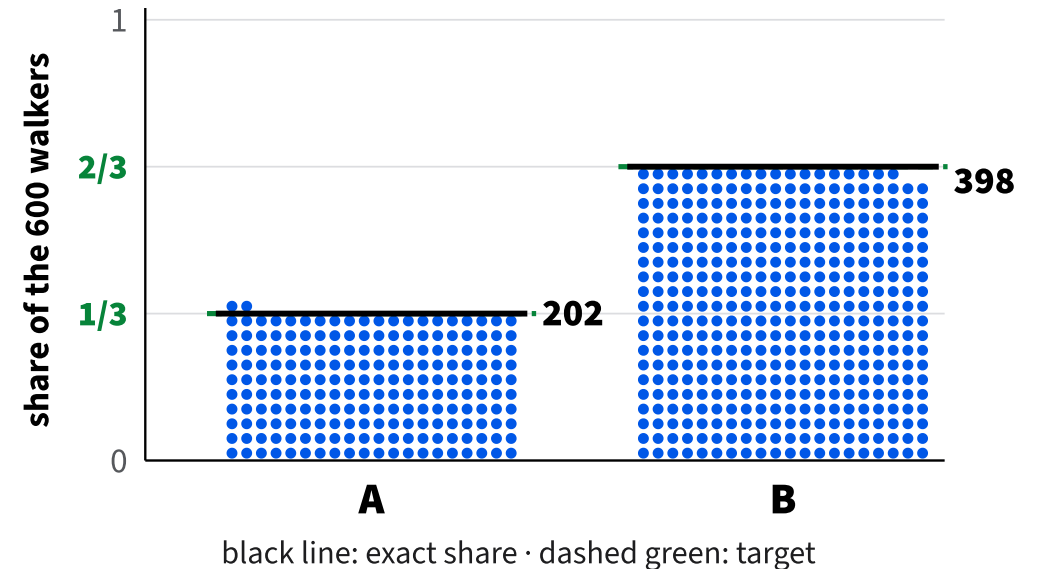
The trick: compare two pictures. Z cancels.

- ▶ **Uphill:** always move.
- ▶ **Downhill:** move sometimes, in proportion to the fit.

Press Run: the walkers settle at $\frac{1}{3}$ and $\frac{2}{3}$, never computing Z .

Metropolis et al., 1953; Hastings, 1970

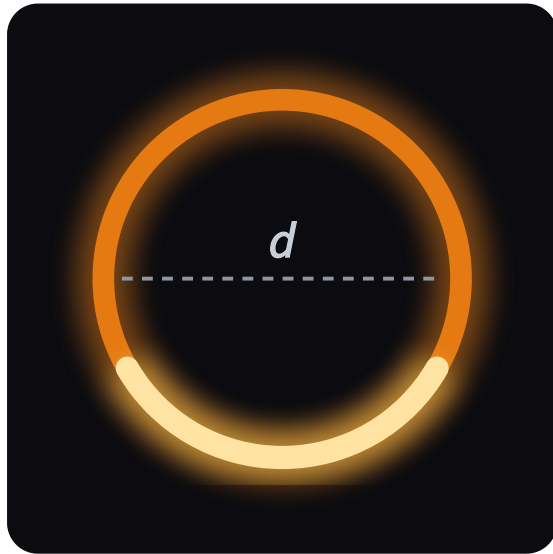
t = 20 from A: always move to B · from B: back with probability $\frac{1}{2}$



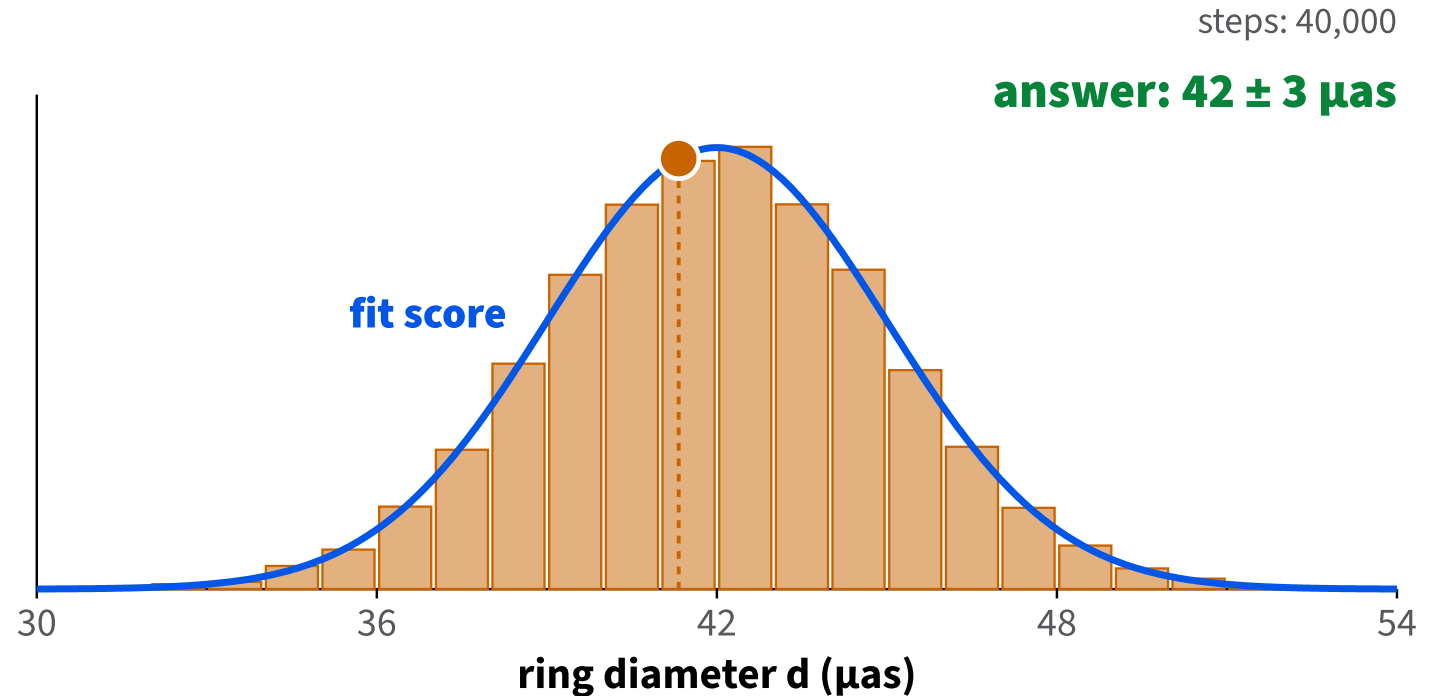
t = 20: 398 of 600 walkers at B, share **0.663** ·
exact $b_{20} = \mathbf{0.6667}$

The walker of Lecture 4

The walker measured the black-hole ring.



$d = 41.3 \mu\text{as}$



The walker moves along the ring's diameter; where it spends its time is the answer.

Result: diameter 42 ± 3 microarcseconds (μas); mass 6.5 billion Suns [Event Horizon Telescope Collaboration, 2019].

Main Question

The walker only compares. Can it also weigh colliding black holes, give neural networks error bars, paint pictures, and help language models reason?

This is indeed the case, after a few changes.

The walker of Lecture 4

Five troubles at the frontier, five fixes.

	Trouble	Fix	Frontier
1	far-apart hills	???	colliding black holes
2	many dimensions	???	COVID-19 lockdowns
3	too much data	???	neural-network error bars
4	no formula	???	image generators
5	all possible texts	???	language-model reasoning

Outline

The walker of Lecture 4

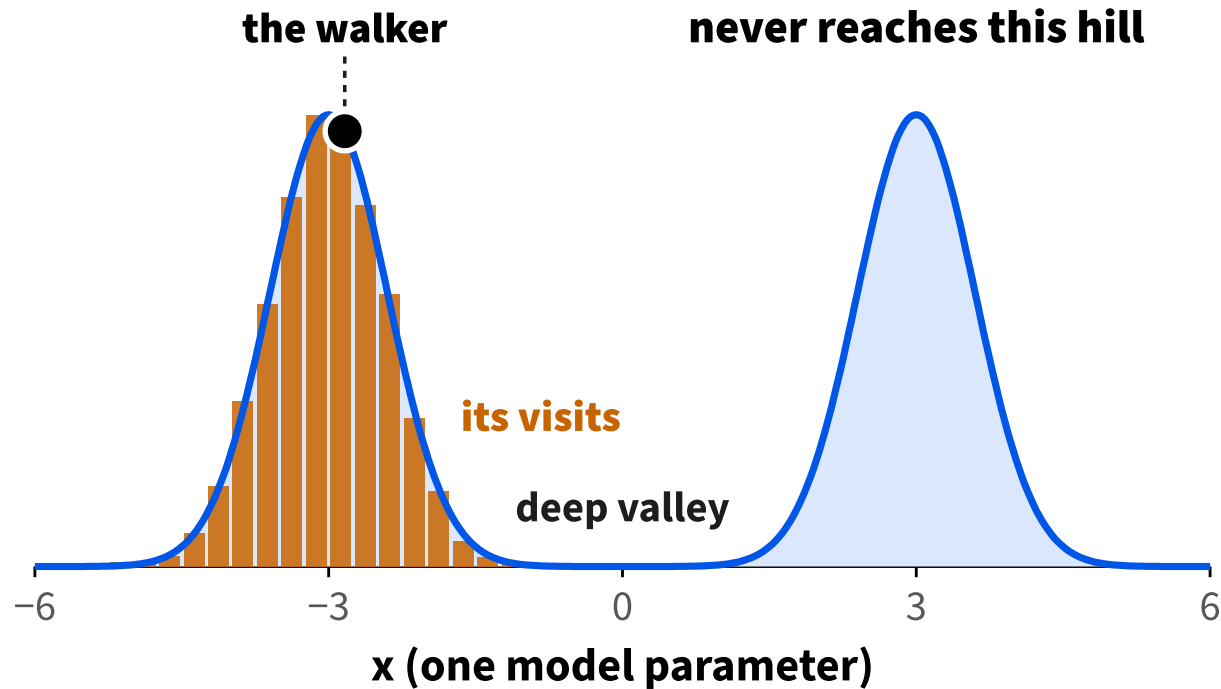
► **Three troubles, three fixes**

The walker as a generator

Conclusion

Trouble 1: far-apart hills

Two far-apart hills trap the walker in one of them.



step **20 000** of 20 000 · walker at $x = -2.84$ · rightmost so far: -0.83

Press Run: in 20 000 steps, the walker never reaches the right hill.

Q: Why is it stuck?

A: Crossing needs many downhill moves in a row: it almost never happens.

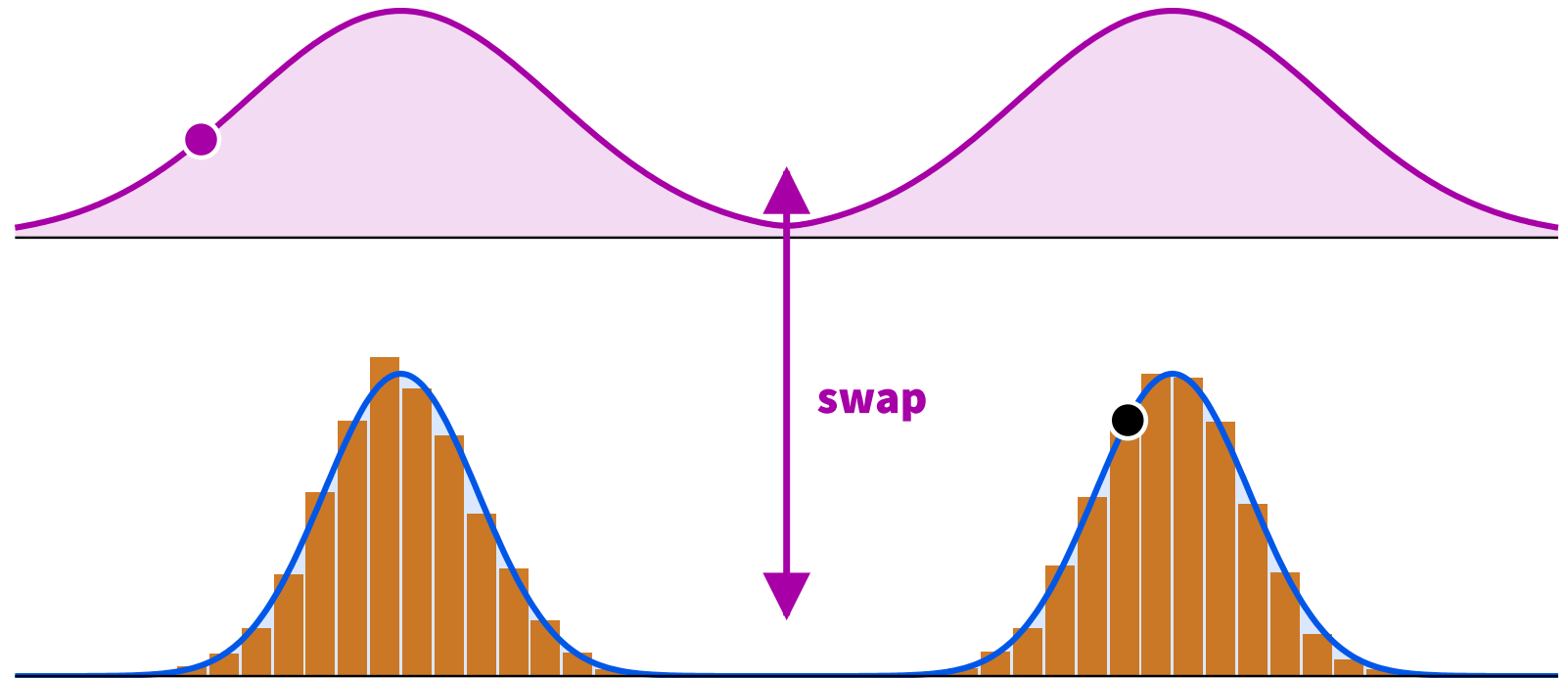
Trouble 1: far-apart hills

Fix: heat a copy until the hills flatten; let it swap with the walker.

hot copy ($T = 4$):
crosses the valley

cold copy ($T = 1$):
the target

cold walker's samples



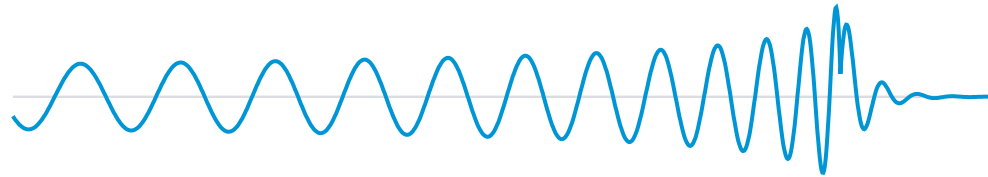
Press Run: the hot copy crosses the valley; after a swap, the cold walker visits both hills.

This is **parallel tempering** [Swendsen and Wang, 1986].

Trouble 1: far-apart hills

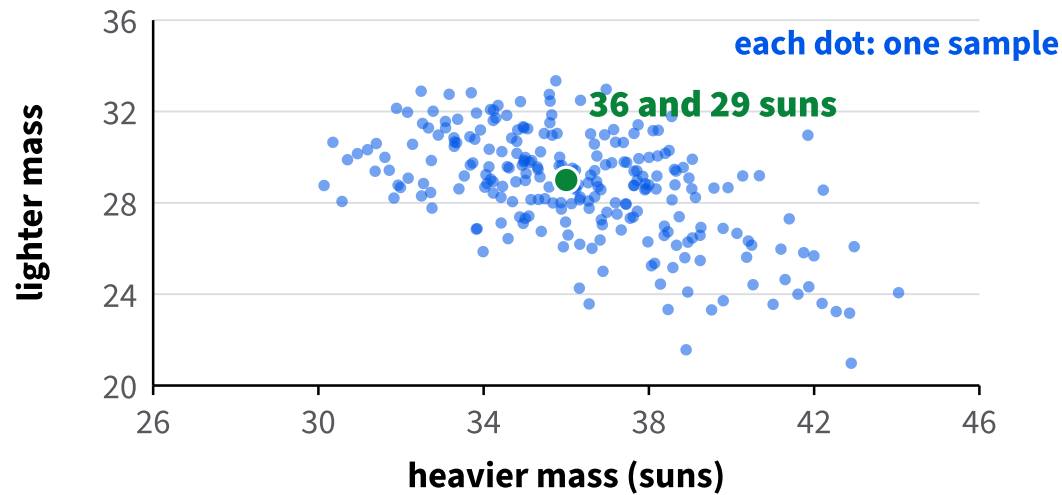
Result: two colliding black holes were weighed this way.

the measured wave (schematic): faster and louder until the merger



time → (about 0.2 s shown)

merger



14 September 2015: a 0.2-second chirp, 1.3 billion light-years away.

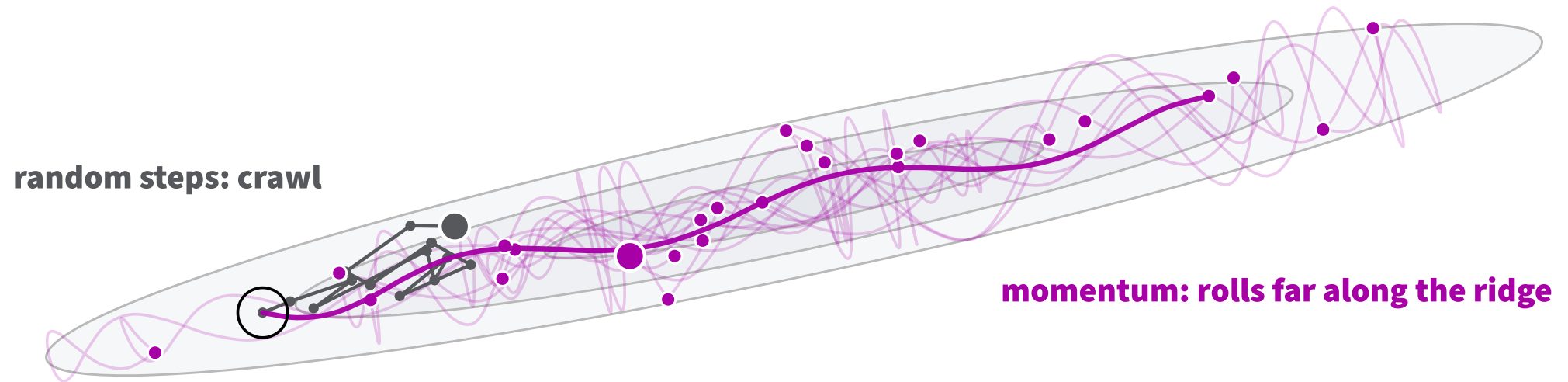
36 and 29 Suns, with error bars.

Veitch et al., 2015; Abbott et al., 2016

The measured wave (schematic) and the walker's samples of the two masses.

Trouble 2: many dimensions

Random steps crawl along a narrow hill. Fix: roll like a ball.

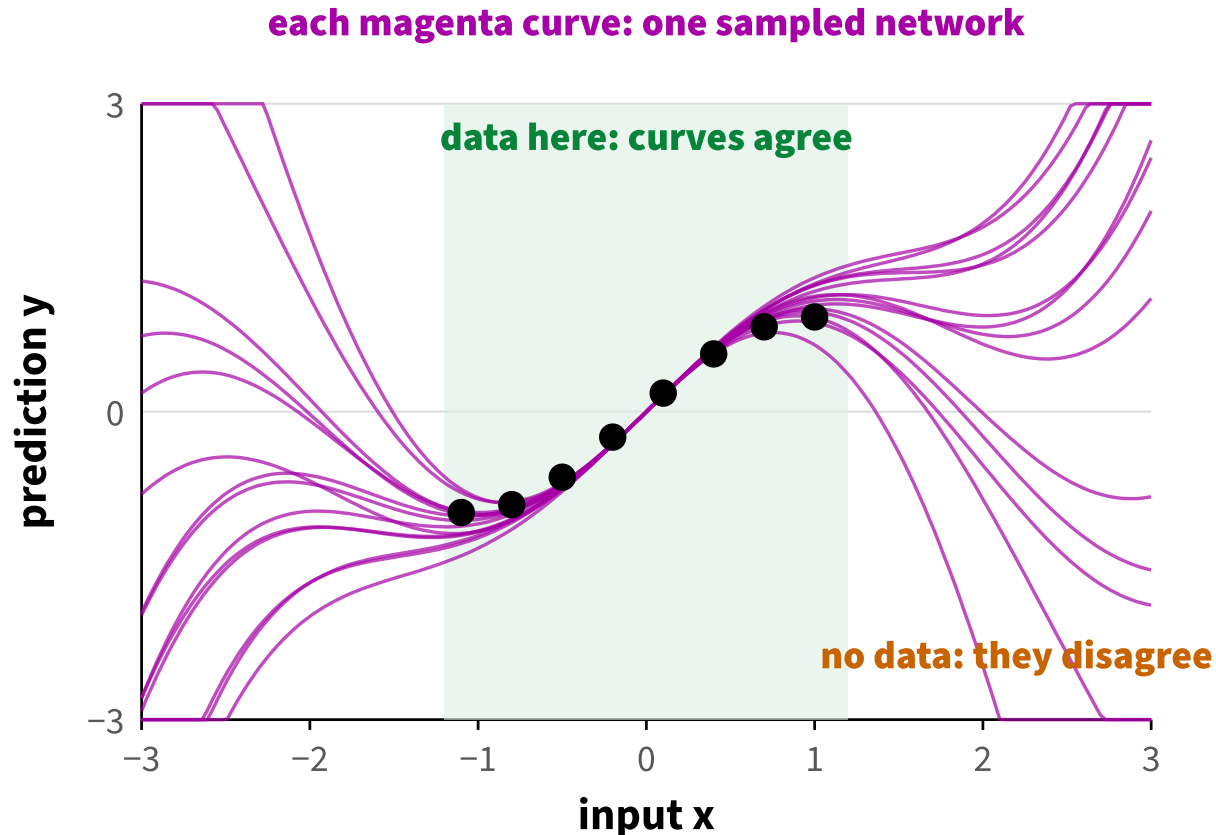


Press Run: grey random steps crawl; the magenta ball keeps its energy and rolls far.

This is **Hamiltonian Monte Carlo** [Duane et al., 1987]; in the Stan software it estimated how lockdowns slowed COVID-19 in 11 countries [Flaxman et al., 2020].

Trouble 3: too much data

Sample a network's weights, and it knows where it is unsure.

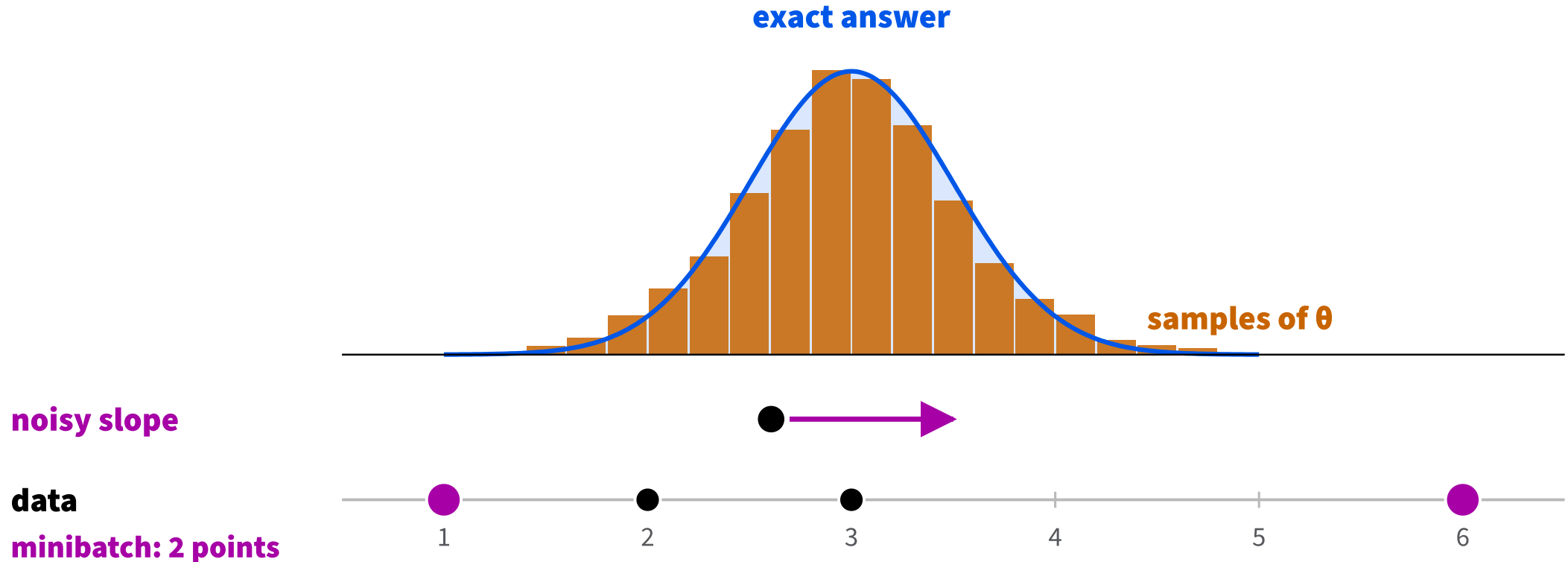


Trouble: one comparison needs **all** the training data, millions of examples.

Each curve is one sampled network: they agree near the data, disagree far away.

Trouble 3: too much data

Fix: step along a noisy minibatch slope, and skip the test.



Press Run: each step uses only two data points, yet the samples match the exact answer.

This is **stochastic-gradient Langevin dynamics** [Welling and Teh, 2011].

Outline

The walker of Lecture 4

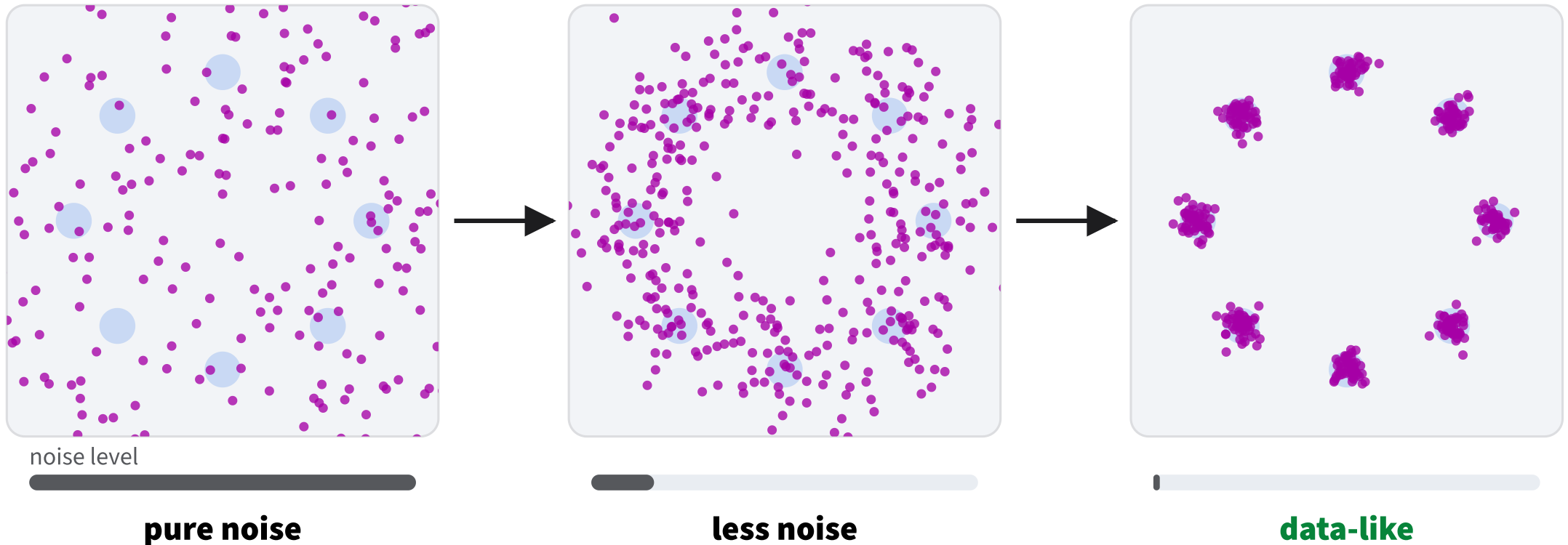
Three troubles, three fixes

► **The walker as a generator**

Conclusion

Trouble 4: no formula

No formula for photos, only examples. Fix: learn the slope, then walk from noise.

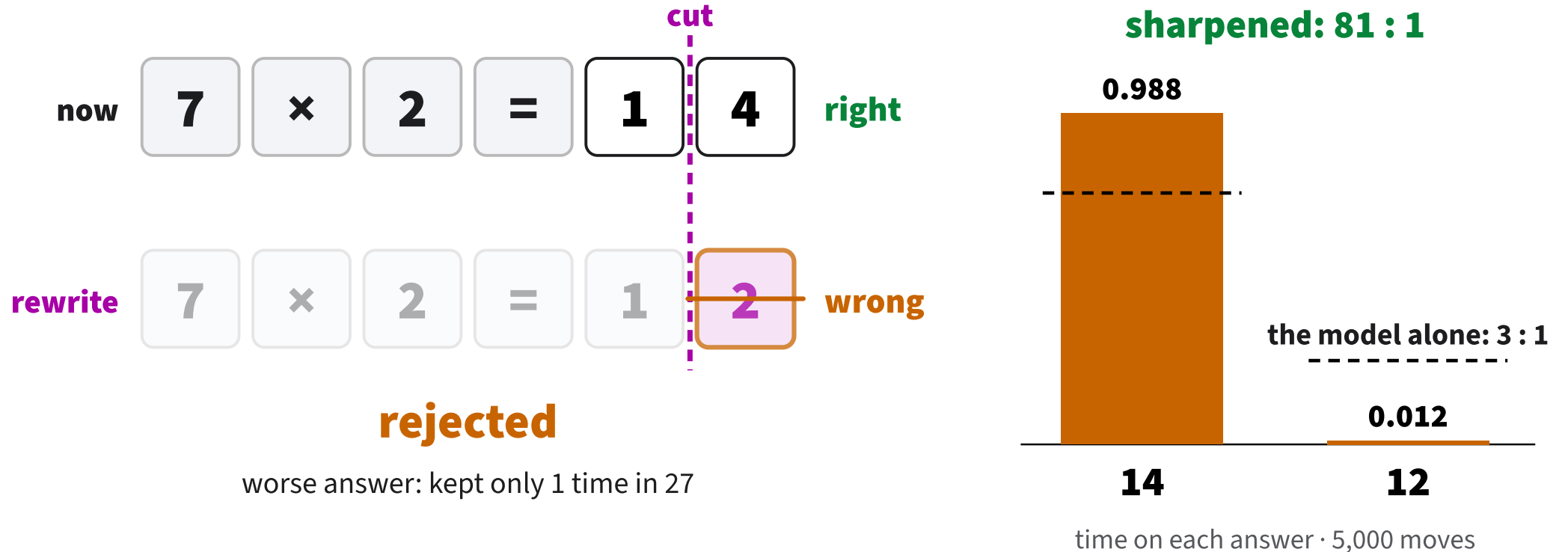


Press Run: points start as noise and walk into the data as the noise shrinks.

This is how **image generators** draw pictures [Song and Ermon, 2019; Ho et al., 2020].

Trouble 5: all possible texts

Sharpen a language model: rewrite the end of an answer, then compare.



Press Run: the walker keeps the right answer far more often than the model alone.

The same model reasons better, with no extra training [Karan and Du, 2025].

Outline

The walker of Lecture 4

Three troubles, three fixes

The walker as a generator

► **Conclusion**

Conclusion

One comparison, five frontiers.

	Trouble	Fix	Frontier
1	far-apart hills	heat a copy, swap	colliding black holes
2	many dimensions	roll like a ball	COVID-19 lockdowns
3	too much data	noisy minibatch slope	neural-network error bars
4	no formula	learn the slope from examples	image generators
5	all possible texts	rewrite the end, compare	language-model reasoning

Conclusion

- ▶ It is possible to sample **without computing Z** : just compare two states.
- ▶ Each frontier changes **one part** of the walk: the copies, the move, or the test.
- ▶ We leave as **open problems**:
 - ▶ Networks that output samples instantly [Dax et al., 2021]: when can we trust them?
 - ▶ Why do "cold" networks predict better [Wenzel et al., 2020]?
 - ▶ One step instead of 1000 for images [Song et al., 2023]?
 - ▶ Rolling like a ball over words, which have no slope?

Selected References I

-  Abbott, B. P., et al. (LIGO Scientific Collaboration and Virgo Collaboration) (2016).
Properties of the binary black hole merger GW150914.
Physical Review Letters, 116(24):241102.
-  Dax, M., Green, S. R., Gair, J., Macke, J. H., Buonanno, A., and Schölkopf, B. (2021).
Real-time gravitational wave science with neural posterior estimation.
Physical Review Letters, 127(24):241103.
-  Duane, S., Kennedy, A. D., Pendleton, B. J., and Roweth, D. (1987).
Hybrid Monte Carlo.
Physics Letters B, 195(2):216–222.
-  Event Horizon Telescope Collaboration (2019).
First M87 Event Horizon Telescope results. VI. The shadow and mass of the central black hole.
The Astrophysical Journal Letters, 875(1):L6.
-  Flaxman, S., Mishra, S., Gandy, A., et al. (2020).
Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe.
Nature, 584:257–261.
-  Hastings, W. K. (1970).
Monte Carlo sampling methods using Markov chains and their applications.
Biometrika, 57(1):97–109.

Selected References II

-  Ho, J., Jain, A., and Abbeel, P. (2020).
Denoising diffusion probabilistic models.
In Advances in Neural Information Processing Systems 33, pages 6840–6851.
-  Karan, A. and Du, Y. (2025).
Reasoning with sampling: your base model is smarter than you think.
arXiv:2510.14901.
-  Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., and Teller, E. (1953).
Equation of state calculations by fast computing machines.
The Journal of Chemical Physics, 21(6):1087–1092.
-  Song, Y. and Ermon, S. (2019).
Generative modeling by estimating gradients of the data distribution.
In Advances in Neural Information Processing Systems 32.
-  Song, Y., Dhariwal, P., Chen, M., and Sutskever, I. (2023).
Consistency models.
In Proceedings of the 40th International Conference on Machine Learning, PMLR 202:32211–32252.
-  Swendsen, R. H. and Wang, J.-S. (1986).
Replica Monte Carlo simulation of spin-glasses.
Physical Review Letters, 57(21):2607–2609.

Selected References III

-  Veitch, J., Raymond, V., Farr, B., et al. (2015).
Parameter estimation for compact binaries with ground-based gravitational-wave observations using the LALInference software library.
Physical Review D, 91(4):042003.
-  Welling, M. and Teh, Y. W. (2011).
Bayesian learning via stochastic gradient Langevin dynamics.
In Proceedings of the 28th International Conference on Machine Learning, pages 681–688.
-  Wenzel, F., Roth, K., Veeling, B. S., Świątkowski, J., Tran, L., Mandt, S., Snoek, J., Salimans, T., Jenatton, R., and Nowozin, S. (2020).
How good is the Bayes posterior in deep neural networks really?
In Proceedings of the 37th International Conference on Machine Learning, PMLR 119:10248–10259.

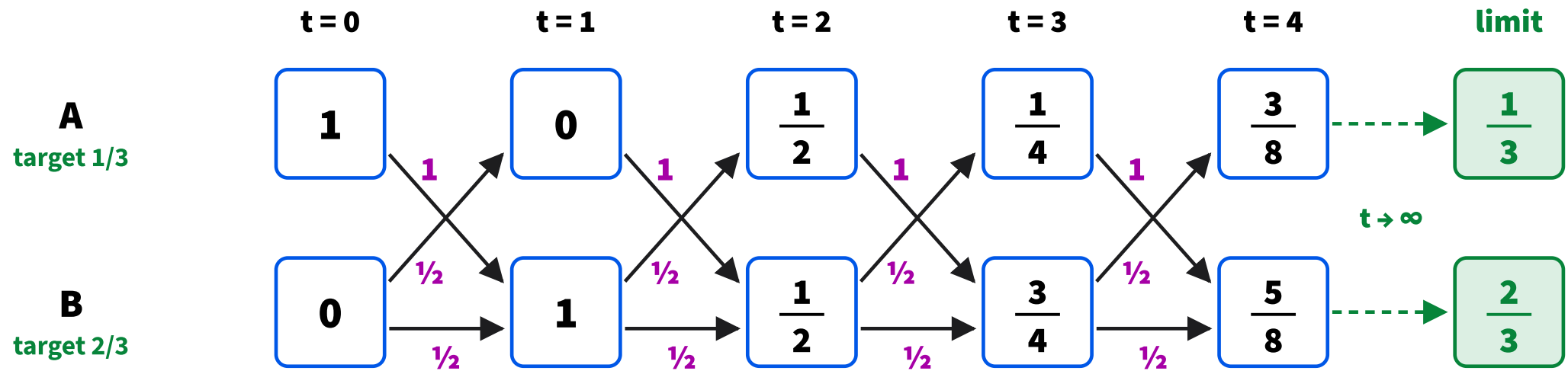
Thank you!

Next: Lab 5, Saturday 10 October: the expectation–maximization algorithm at the frontier

Backup: the walker of Lecture 4

Why the walkers settle at $\frac{1}{3}$ and $\frac{2}{3}$: the flows balance.

From A, a walker always moves to B; from B, it moves back half the time. At $\frac{1}{3}$ and $\frac{2}{3}$, as many walkers go each way: $\frac{1}{3} \cdot 1 = \frac{2}{3} \cdot \frac{1}{2}$ (Lecture 4, Theorem 9).



each box = the sum, over the arrows into it, of (box $\times$ weight): e.g. $\frac{3}{4} = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2}$

Each box is the share at A or B after t steps; each arrow carries a move probability.

Backup: the walker of Lecture 4

The whole walker is a few lines of Python, and it never uses Z .

```
import random
random.seed(0)
g = {"A": 1, "B": 2}                # fit scores; Z is never computed
x, visits = "A", {"A": 0, "B": 0}
for t in range(100_000):
    y = "B" if x == "A" else "A"    # propose the other picture
    if random.random() < min(1, g[y] / g[x]): # uphill: always; downhill: g(new)/g(old)
        x = y
    visits[x] += 1
print({s: v / 100_000 for s, v in visits.items()})
{'A': 0.33392, 'B': 0.66608}
```

The walker spent 0.334 of its time at A and 0.666 at B, close to $\frac{1}{3}$ and $\frac{2}{3}$.

Backup: Trouble 1

The swap of parallel tempering is again a comparison.

A copy at temperature T walks on $g(x)^{1/T}$. Two copies at $T_1 < T_2$ propose to swap their states x_1, x_2 ; the test compares the swapped pair with the current pair:

$$\frac{g(x_2)^{1/T_1} \cdot g(x_1)^{1/T_2}}{g(x_1)^{1/T_1} \cdot g(x_2)^{1/T_2}} = \left(\frac{g(x_2)}{g(x_1)} \right)^{\frac{1}{T_1} - \frac{1}{T_2}}.$$

Accept with $\min\{1, \text{this ratio}\}$. With $T_1 = 1$ and $T_2 = 4$, the power is $\frac{3}{4}$. No normalizer appears.

Backup: Trouble 2

One roll, step by step (the leapfrog method).

Height $U(x) = -\ln g(x) = \frac{1}{2} \cdot x^2$, slope $U'(x) = x$. Start at $x = 1$ with momentum $v = 1$, step 0.5; energy $H = U(x) + \frac{1}{2} \cdot v^2 = 1$.

$$v = 1 - 0.25 \cdot 1 = 0.75$$

$$x = 1 + 0.5 \cdot 0.75 = 1.375$$

$$v = 0.75 - 0.25 \cdot 1.375 = 0.406$$

$$H' = \frac{1}{2} \cdot 1.375^2 + \frac{1}{2} \cdot 0.406^2 = 1.028$$

$$\min\{1, e^{-(1.028-1)}\} = 0.97$$

half kick

drift

half kick

energy almost kept

the usual test